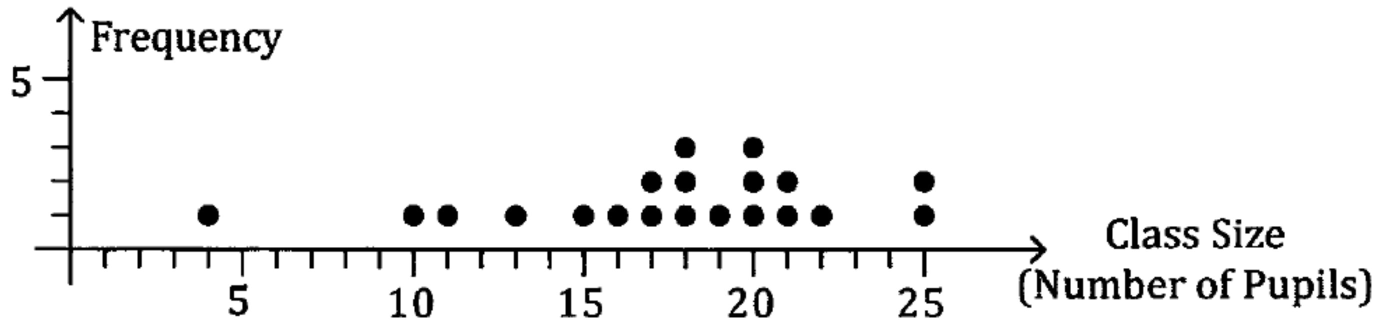
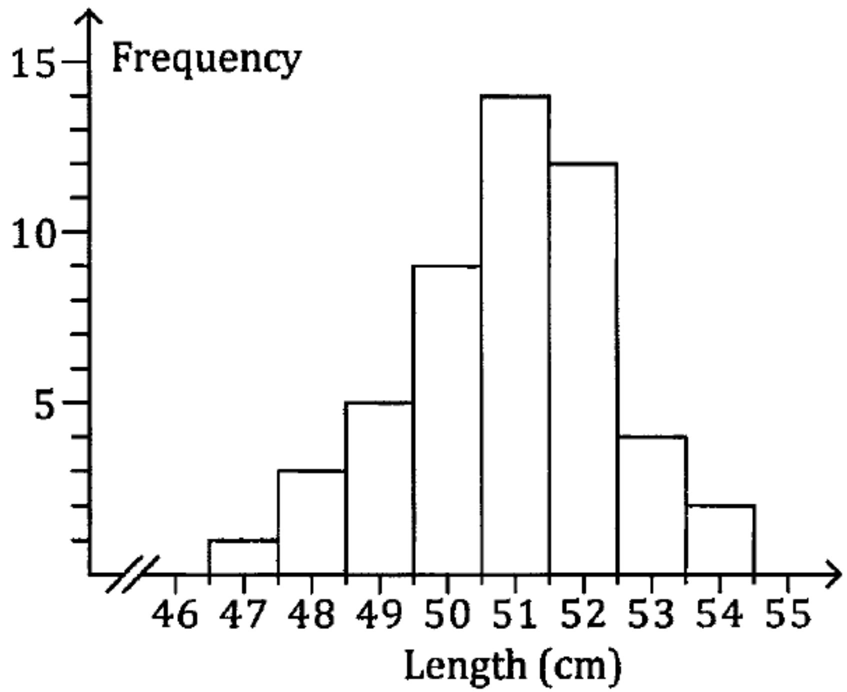
Displaying Data

How do we display numerical data?

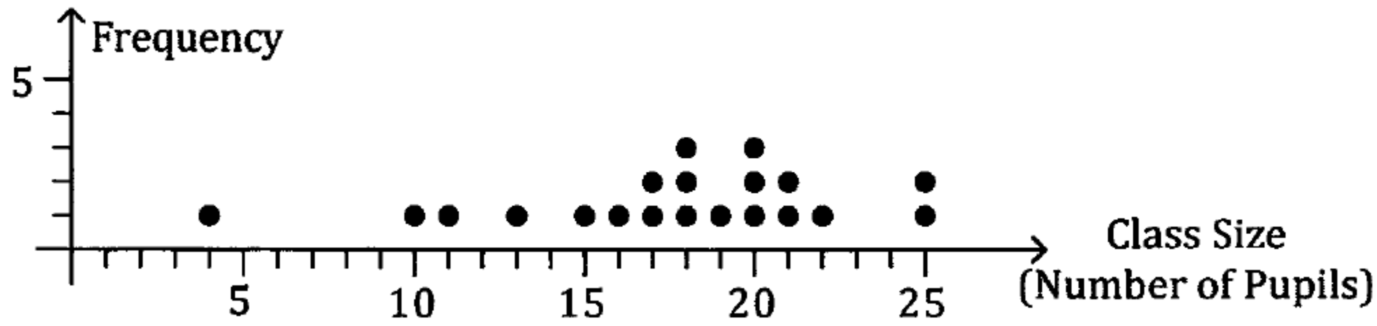


Dot frequency diagram



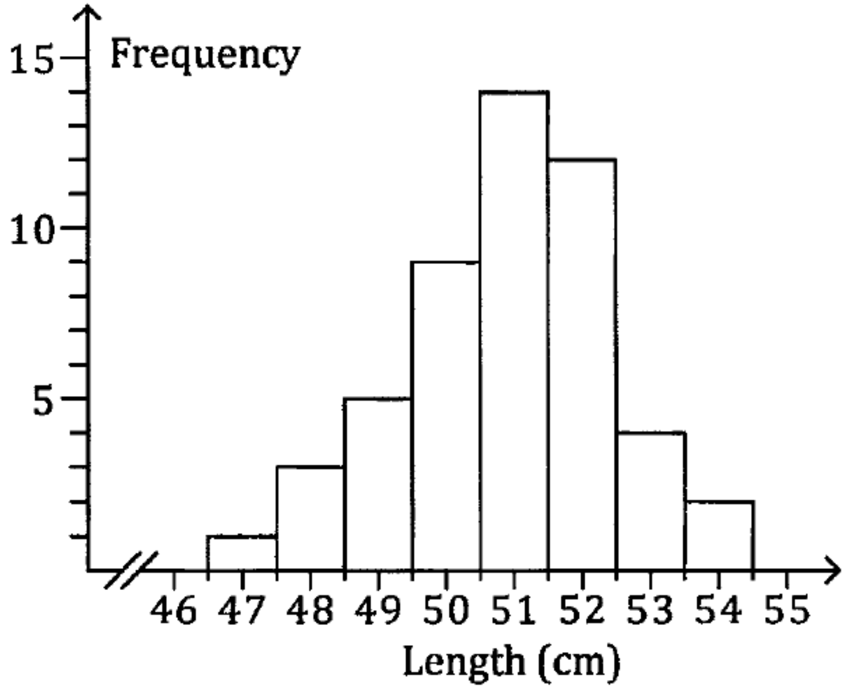
Frequency histogram

Dot frequency diagram



- Discrete data – number of pupils, not continuous.
- y-axis is always frequency
- x-axis is always numbers (in this case, integer)

Frequency histogram



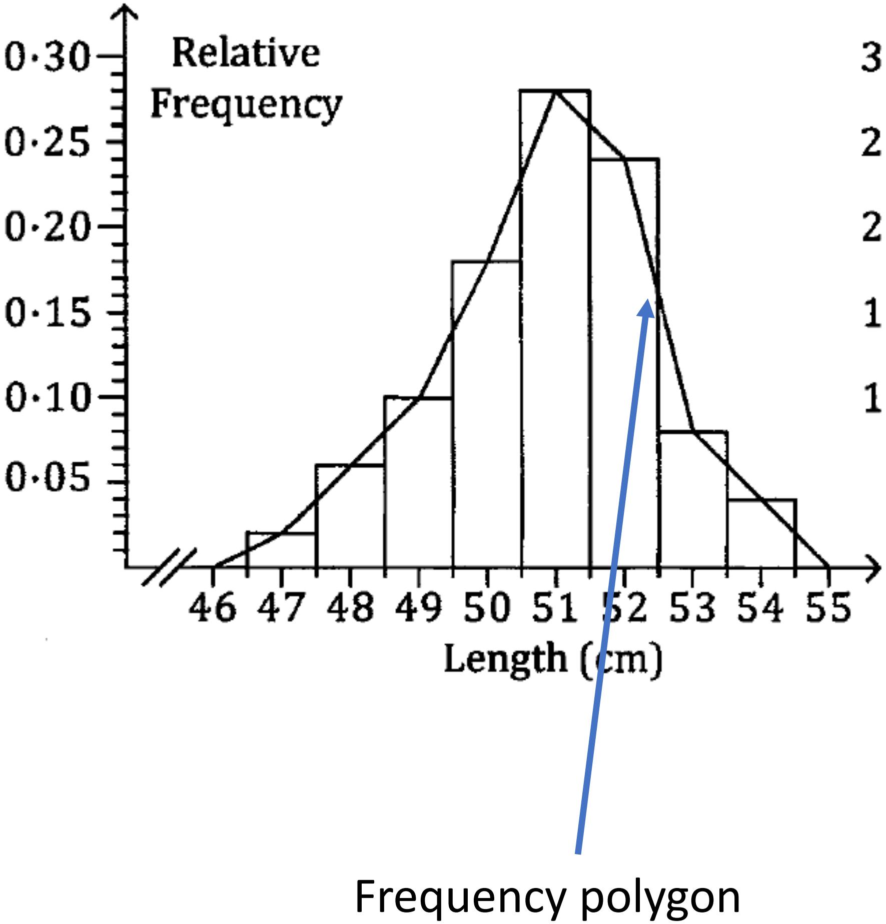
- Length – continuous data
- 50 cm – between 49.5 to 50.5

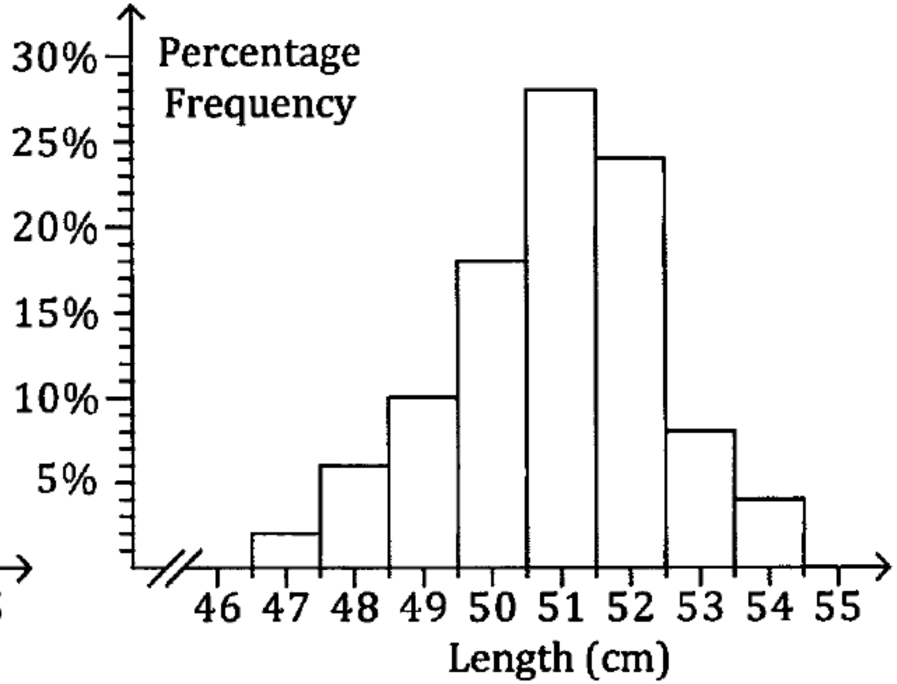
Length (cm)	47	48	49	50	51	52	53	54
Frequency	1	3	5	9	14	12	4	2

Frequency histogram

- No gaps between the bar except when frequency = 0.
- Usually have 6 – 10 equal intervals.
- y-axis is always frequency
- x-axis is always numbers
- The frequency can be expressed in
  - Relative frequency – fraction of the whole
  - Percentage frequency – percentage of the whole

Length (cm)	47	48	49	50	51	52	53	54
Frequency	1	3	5	9	14	12	4	2
Relative Frequency	0·02	0·06	0·10	0·18	0·28	0·24	0·08	0·04
Percentage Frequency	2%	6%	10%	18%	28%	24%	8%	4%





Worksheet

The following data give the number of hours 60 self-employed brick layers worked during a particular week.

12	21	31	31	40	48	46	50	52	42	47	32
44	45	62	51	23	34	36	30	43	45	48	26
56	52	45	42	42	36	37	31	32	41	18	48
25	47	43	36	38	39	45	58	53	36	38	31
49	31	47	52	30	40	39	43	44	33	34	42

a. Represent the data on a frequency distribution table.

b. Draw a histogram of the data.

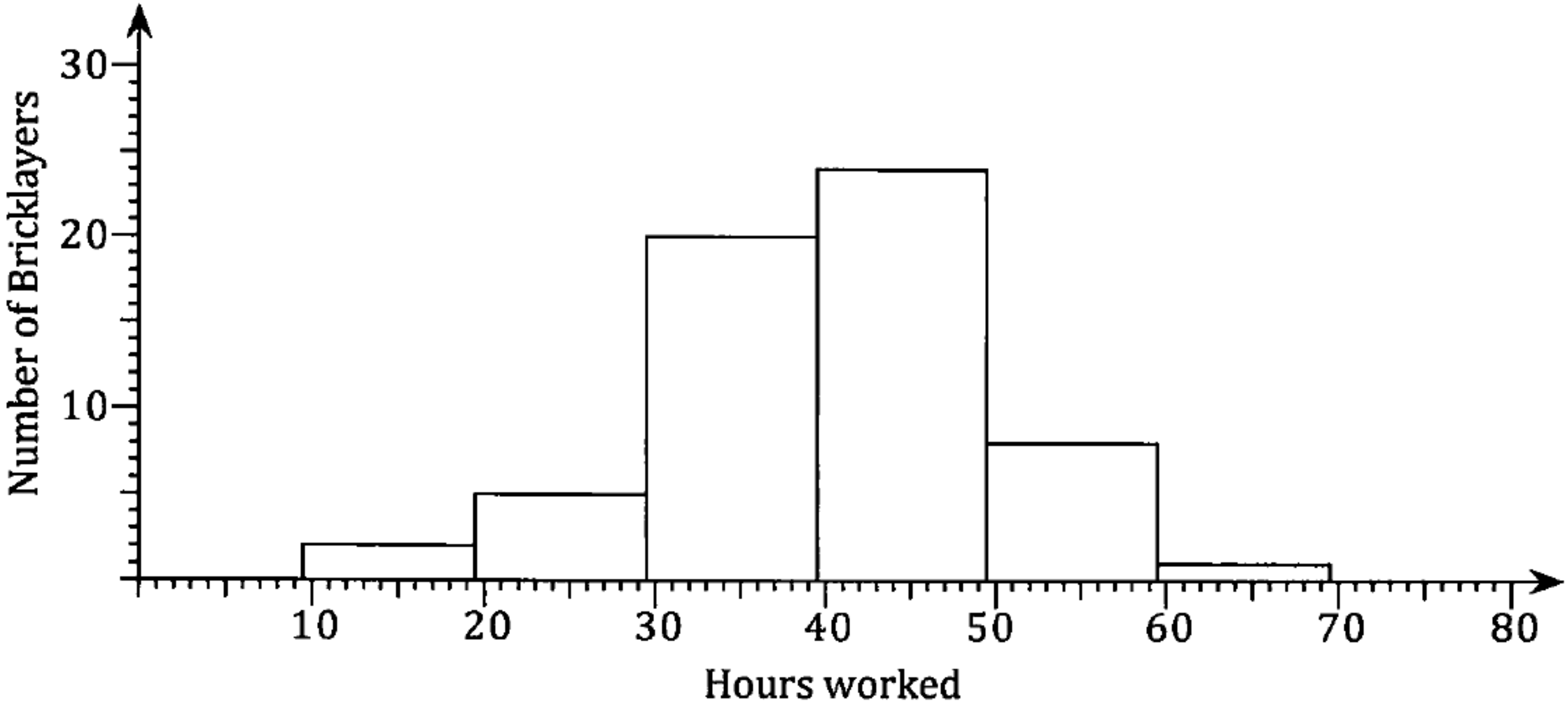
Determine the interval

12	21	31	31	40	48	46	50	52	42	47	32
44	45	62	51	23	34	36	30	43	45	48	26
56	52	45	42	42	36	37	31	32	41	18	48
25	47	43	36	38	39	45	58	53	36	38	31
49	31	47	52	30	40	39	43	44	33	34	42

Minimum = 12 hours

Maximum = 62 hours

No of hours	Tally	Frequency
10 - 19	||	
20 - 29	~~||||~~	
30 - 39	~~||||~~ ~~||||~~ ~~||||~~ ~~||||~~	
40 - 49	~~||||~~ ~~||||~~ ~~||||~~ ~~||||~~ || ||	
50 - 59	~~||||~~ |||	
60 - 69	|	



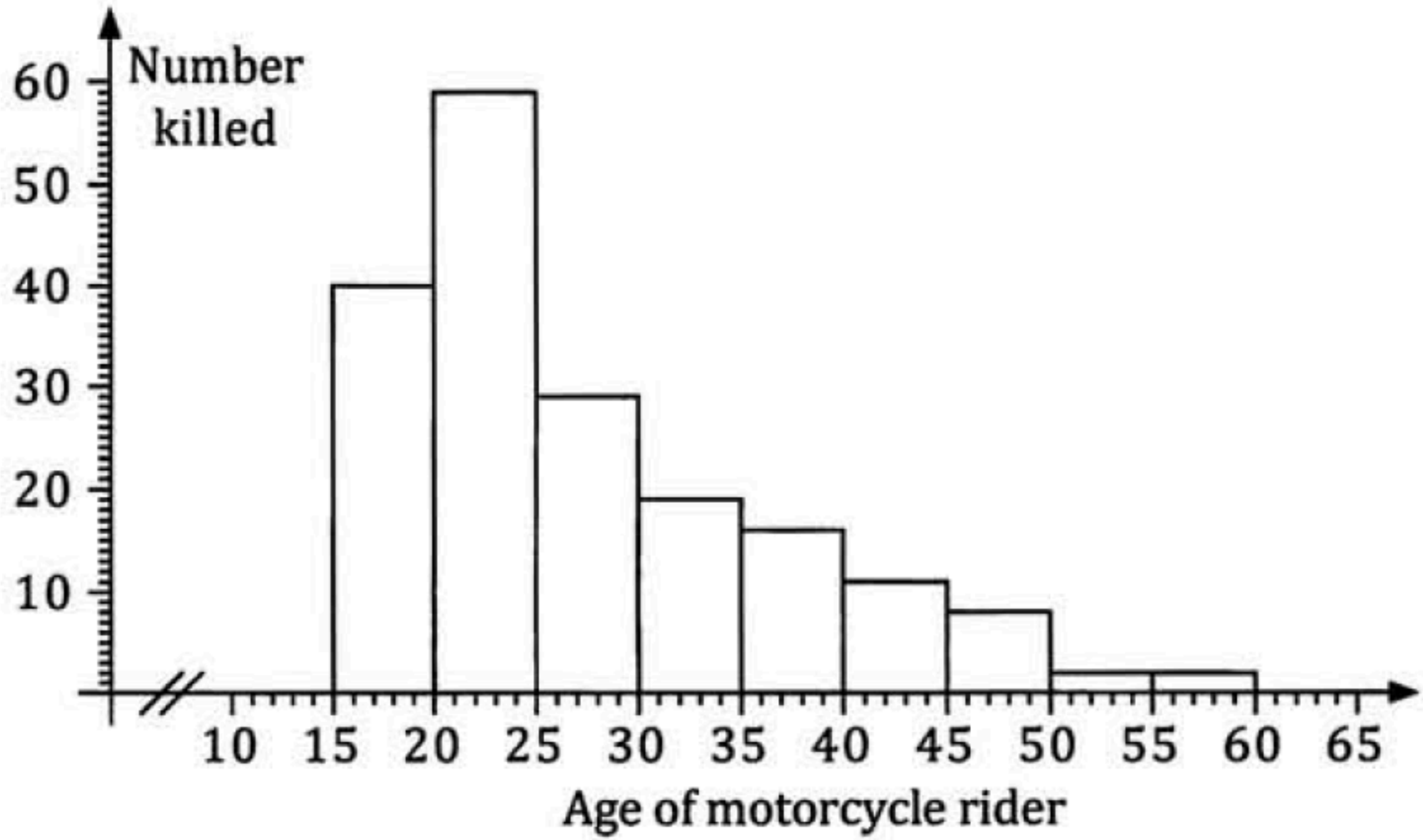
30
20
10
Number of Bricklayers
10
20
30
40
50
60
70
80
Hours worked

Example 2.

The road accident statistics for a country for one year showed that for motorcyclists (drivers not passengers) in the age range fifteen to fifty-nine, 186 had died in road accidents with the distribution of the ages of these riders as shown on the right,

Display this information as a frequency histogram.

Age (x yrs)	Drivers Killed
$15 \leq x < 20$	40
$20 \leq x < 25$	59
$25 \leq x < 30$	29
$30 \leq x < 35$	19
$35 \leq x < 40$	16
$40 \leq x < 45$	11
$45 \leq x < 50$	8
$50 \leq x < 55$	2
$55 \leq x < 60$	2


Number
killed
60
50
40
30
20
10
10
15
20
25
30
35
40
45
50
55
60
65
Age of motorcycle rider